

TUTORIAL – 6: Manual Test Execution

Aim

To execute manual test cases using multiple input conditions, compare expected and actual results, identify defects if any, and prepare a detailed **Test Execution Report** and **Test Summary Report**.

Description

Manual Testing is the process of testing software **without using any automation tools**. In manual testing, the tester executes test cases manually, enters different input values, compares the **Expected Result** with the **Actual Result**, and records whether the test case **Passes or Fails**.

The objective of manual test execution is to verify that the application works correctly under different input conditions.

During manual testing, the tester performs:

- Positive Testing (Valid Inputs)
- Negative Testing (Invalid Inputs)
- Boundary Testing
- Validation Testing

After execution, the tester prepares:

- Test Execution Report
- Defect Report (if defects are found)
- Test Summary Report

Problem Statement

Execute manual test cases for the **Login Module** of the **Student Management System** using different input conditions. Record the execution results, identify any defects, and prepare a Test Summary Report.

How to Perform This Practical (Student Procedure)

Step 1: Study the Requirements

Read the Software Requirement Specification (SRS) and understand the login functionality.

Requirements:

- User must enter a valid username and password.
- Invalid credentials should display an error message.
- Blank fields should show validation messages.

Step 2: Prepare Test Cases

Create test cases using different input conditions.

Test Type	Example
Positive Test	admin / admin123
Negative Test	admin / 12345
Blank Input	"" / ""
Invalid Username	student / admin123
Invalid Password	admin / xyz123

Step 3: Run the Application

Open the application in **VS Code**, **PyCharm**, or a web browser.

Step 4: Execute Test Cases

Run each test case manually.

For each test case:

- Enter the input.
- Click **Login**.
- Observe the result.
- Compare the **Expected Result** with the **Actual Result**.
- Mark the test case as **Pass** or **Fail**.

Step 5: Record the Results

Prepare a Test Execution Report with:

- Test Case ID
- Input
- Expected Result
- Actual Result
- Status

Step 6: Prepare Test Summary Report

Count:

- Total Test Cases

- Passed Test Cases
- Failed Test Cases
- Defects Found

Implementation Details

The Student Management System allows registered users to log in using valid credentials. Manual testing is performed using different input conditions to verify system behavior.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	PyCharm / Visual Studio Code
Input/Test Data	admin / admin123, admin / 12345, blank values

Source Code

```
username = input("Enter Username: ")
password = input("Enter Password: ")
```

```
if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	User Login Authentication
Test Scenario	Verify login with valid credentials
Test Objective	Ensure only authorized users can log in
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is running and Login page is open
Postconditions	User is redirected to Home Page
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome / Windows 10
Test Data	Username: admin Password: admin123
Expected Result	Login Successful
Actual Result	Login Successful
Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Manual Test Execution

Test Case ID	Input	Expected Result	Actual Result	Status
TC001	admin / admin123	Login Successful	Login Successful	Pass
TC002	admin / 12345	Invalid Password Message	Invalid Password Message	Pass
TC003	student / admin123	Invalid Username Message	Invalid Username Message	Pass
TC004	Blank Username	Username Required	Username Required	Pass
TC005	Blank Password	Password Required	Password Required	Pass
TC006	Blank Username & Password	Validation Messages	Validation Messages	Pass

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Application	-	Login page displayed	Login page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Login Successful	Login Successful	Pass
5	Enter Wrong Password	12345	Error Message	Error Message Displayed	Pass

6	Leave Fields Blank	""	Validation Message	Validation Message Displayed	Pass
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Test Summary Report

Item	Count
Total Test Cases	6
Passed	6
Failed	0
Defects Found	0
Pass Percentage	100%

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Study SRS	Requirement Document
2	Design	Test Planning	Prepare Test Plan	Test Plan
3	Development	Test Case Design	Create Manual Test Cases	Test Case Document
4	Testing	Test Execution	Execute Test Cases	Test Execution Report
5	Testing	Defect Reporting	Record Defects (if any)	Bug Report
6	Testing	Test Closure	Prepare Test Summary Report	Test Summary Report

Observation

Sr. No.	Observation	Remarks
1	Login successful with valid credentials	Pass
2	Invalid credentials rejected correctly	Pass
3	Blank fields display validation messages	Pass
4	Manual test cases executed successfully	Completed
5	Test Summary Report prepared	Completed

Output

Manual Test Execution Completed

Total Test Cases : 6
Passed : 6
Failed : 0
Pass Percentage : 100%

Result:

All Manual Test Cases Executed Successfully

Solve the Below Mention Questionnaires

1. What is Manual Testing?

A) Testing using automation tools

B) Testing performed manually without automation tools

C) Database Testing

D) Network Testing

Answer: B) Testing performed manually without automation tools

2. Which document records actual results of executed test cases?

- A) Design Document
- B) Test Execution Report**
- C) User Manual
- D) RTM

Answer: B) Test Execution Report

3. What is compared during test execution?

- A) Source Code and Design Document
- B) Expected Result and Actual Result**
- C) Database and UI
- D) Hardware and Software

Answer: B) Expected Result and Actual Result

4. Which input condition checks validation handling?

- A) Valid Username and Password
- B) Blank Input Values**
- C) Correct Login Credentials
- D) Successful Transaction

Answer: B) Blank Input Values

5. What is the purpose of a Test Summary Report?

- A) Writing Source Code
- B) Designing Database Tables
- C) Providing Overall Testing Status and Results**
- D) Creating User Interfaces

Answer: C) Providing Overall Testing Status and Results

Conclusion / Outcome

- Understood the complete process of **Manual Test Execution**.
- Executed manual test cases using multiple input conditions, including valid, invalid, and blank inputs.
- Compared expected results with actual results and recorded the execution status.
- Prepared a **Test Execution Report** and **Test Summary Report**.
- Learned how manual testing helps verify software functionality and identify defects before software release.